Gender norms and how they may shift*

[redacted]

Gender norms come in many shapes and forms. They provide expectations on how men and women should act in a society. We use an overlapping generations model to study how gender norms shape the behavior of women. If there is large enough social pressure, women partially comply with the norm set by society. The norm, however, will not be sustained over time because forward-looking individuals anticipate how their actions affect future generations with which they interact. Furthermore, we show that there is a unique Markov-perfect equilibrium in affine strategies, in which behavior converges to a steady state in which the social norm is the average populations preferred action.

Keywords: *Gender norms, overlapping generations, Markov-perfect equilibrium*

1 Introduction

Traditionally, women are seen as the care-taker of the family while men are seen as the breadwinner. This idea would also translate into the gender norm where mothers are almost expected to be the care-taker of the family especially after giving birth to a child. Hence, for mothers who would want to re-enter the labor market, they might face social pressure by going against this expectation. Therefore, the players in this paper would be mothers within a society each having a type of either 0 or 1 (tradition, non-traditional). Their action would be on a spectrum of [0,1] where 0 would be choosing to take care of the family completely

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after giving birth, and 1 would be choosing to work completely. Because the gender norm is seen as the average action amongst all players for the generation they are alive in, because the actions are those of the mothers, this also implies that gender norm would be constituted by the action of other women within the society. And because social pressure is described as a player's belief on how important the gender norm is, it would also mean that social pressure would be coming from the action of other mothers in the society.

By studying how gender norms may potentially keep mothers' from re-entering the labor market, it could give us insights on how to encourage a change in such situation. Although the model only studies how the gender norm would impact women, the gender norm set on women could potentially harm both men and women. It harms women in the way that this expectation for women to take care of the family may hinder their career development. Furthermore, if men are expected to provide for the entire family, the financial burden would be on one person whereas if both parties worked, the financial pressure to support for the family can be shared by both the mother and the father. Therefore, by looking at potential ways to encourage mother's re-entry into the labor force might be helpful not only for women but also for the well-being of the family.

The idea of how social norms would impact the action of players has been studied using different models in economics. In general, social norm would influence people's actions and a change in the people's action would also change the existing social norm. Azar (2004) studied how the social norm of tipping has sustained through a model where the motivation for tipping is not only to follow the social norms but also to send a positive signal which may benefit the player tipping. They showed that if players in the population are of the same type, then the current social norm would erode to the social norm most suited to the player's type. Furthermore, they also showed that if the current social norm has not changed over time, then this sustainment is done through players deriving benefits from this action. Social norms might also change due to the action of a specific permanent player in the population; see Acemoglu and Jackson (2014). They employed an overlapping generations model where

each player could either choose a high or low action. Depending on how history (signal from the permanent player) is interpreted by the players, they would choose an action corresponding to what they perceive the social norm to be. Social norms could also be sustained due to potential punishments from deviation to it, and due to the fear of punishments, players would comply to the social norm; see Axelrod (2006). However, they have also shown that a new norm would be formed overtime as the fear to punishment subdues. It is also possible that social norm would hinder social change through the inclusion of social inertia de Mesquita and Shadmehr (2023). They have shown that with the inclusion of social inertia, it would mix up the signal on how an individual believes others would act. And revealing more information would help individuals change their actions.

Although social norms and gender norms may be similar at first sight, or that gender norms might be a subset of social norms, in the situation where it concerns a mother's re-entry to the labor market after birth, having one prominent player choose an action different to the current gender norm may not be enough for the situation to change. As gender norms are seen as a product through other people's action within the society and a change in gender norm would require a shift in power (Cislaghi and Heise (2020)), the encouragement to change the gender norm would require a more general change in the entire population rather than a single player's action. The gender norm in the model is hence described as the average action amongst all players in the generations for which they are alive in.

2 Model

We consider an overlapping generations model to explain how social norms may persist or change over time. In each generation, a continuum of individuals with mass 1/2 is born that live for 2 generations so that in each generation, there is a total mass 1 of individuals. Every individual i chooses an action $a_i \in [0, 1]$, indicating how closely individual i follows the traditional gender norm. Action 1 indicates complete adherence to the traditional gender norm, whereas action 0 indicates acting against the gender norm. For example, the action

taken could represent the choice of a career. Each individual i has a type $\theta_i \in \Theta$, representing the preferred action of individual i absent of any social pressure to follow the established norms. Types are drawn independently across individuals and generations. For now, let us assume that types are binary, i.e., $\Theta \in \{0, 1\}$, with a fraction μ of individuals in generation t being of type 1. Each individual takes an action only when they are born, but they derive a utility from both generations in which they live. Let us denote by $\bar{a}_t$ the average action taken by individuals born in generation t. The social norm $b_t = (\bar{a}_{t-1} + \bar{a}_t)/2$ in generation t is the average action taken by any individual alive at time t. If individual i of type θ_i is alive at time t, they derive a utility in period t of

$$u_i(\theta_i, a_i, b_t) = -(a_i - \theta_i)^2 - \lambda(a_i - b_t)^2. \tag{1}$$

The first term in (1) is individual i's disutility of deviating from their preferred action θ_i and the second term is the disutility of deviating from the social norm. Parameter $\lambda \in [0, \infty)$ is the strength of the social norm. Individual i born in period t receives a lifetime utility of

$$U_i(\theta_i, a) = u_i(\theta_i, a_i, b_t) + \delta u_i(\theta_i, a_i, b_{t+1}),$$

where $\delta \in [0, 1]$ is the inter-generational discount factor. There is anonymity in the large: individuals observe population actions chosen in previous generations, but not the actions chosen by separate individuals. That is, the history at time t is $h_t = (\bar{a}_0, \bar{a}_1, \ldots, \bar{a}_{t-1})$. A *strategy* of player i born in generation t is a map σ_i that assigns to each (θ_i, h_t) an action $\sigma_i(\theta_i, h_t) \in [0, 1]$. A *Markov strategy* of player i is a strategy σ_i that depends on the entire history only through the average population action $\bar{a}_{t-1}$ in the previous generation. We look for symmetric subgame-perfect equilibria in Markov strategies. Symmetry imposes that each player i uses the same strategy σ as a map from histories to actions.

The exact law of large numbers implies that in any symmetric strategy profile σ, exactly a fraction μ of individuals born in generation t chooses action $\sigma(1, \bar{a}_{t-1})$ and a fraction $1 - \mu$ chooses action $\sigma(0, \bar{a}_{t-1})$. Given the population action $\bar{a}_{t-1}$, the social norm in period t is

thus

$$b_t = \frac{1}{2}\Big(\bar{a}_{t-1} + \mu\sigma(1, \bar{a}_{t-1}) + (1-\mu)\sigma(0, \bar{a}_{t-1})\Big). \tag{2}$$

3 Main result

For the sake of tractability, we will want to place additional restrictions beyond the Markov property on the class of strategies that we study. We begin this section by motivating those. A natural subclass of strategies to study is those that are *monotone* in $\bar{a}_{t-1}$: the closer the average population action from the last generation is to your preferred action θ_i, the closer will your action be to your preferred action. Monotone strategies converge in the following sense.

Lemma 3.1. *The average population action $\bar{a}_t$ in any symmetric monotone Markov strategy converges to μ as $t \to \infty$.*

Proof. Fix any symmetric monotone Markov strategy σ. Denote by

$$\bar{a}_t(\bar{a}_{t-1}) = \mu\sigma(1, \bar{a}_{t-1}) + (1-\mu)\sigma(0, \bar{a}_{t-1}) \tag{3}$$

the implied average population action in generation t, given $\bar{a}_{t-1}$. It follows from monotonicity of σ that $\bar{a}_t$ is monotone. Fix an arbitrary $\bar{a}_0 \in [0,1]$. If $\bar{a}_t(\bar{a}_0) = \bar{a}_0$, then the sequence $(\bar{a}_t)_{t\geq 0}$ is constant and, hence, trivially converges. Suppose next that $\bar{a}_t(\bar{a}_0) > \bar{a}_0$. Montonicity implies $\bar{a}_2 = \bar{a}_t(\bar{a}_1) > \bar{a}_t(\bar{a}_0) = \bar{a}_1$. An iterative argument shows that $(\bar{a}_t)_{t\geq 0}$ is monotonically increasing. Because it is bounded above by 1, it must converge. The case $\bar{a}_t(\bar{a}_0) < \bar{a}_0$ is analogous. We conclude that $(\bar{a}_t)_{t\geq 0}$ converges to some limit $\bar{a}_\infty$.

In the limit, we must have $b_\infty = \bar{a}_\infty$, hence each individual i chooses a_i that maximizes

$$U_i(\theta_i, a) = -(1+\delta)(a_i - \theta_i)^2 - (1+\delta)\lambda(a_i - \bar{a}_\infty)^2.$$

Since there is a continuum of individuals in each generation, individual i has no influence on the average population action $\bar{a}_\infty$. Therefore, the first-order necessary condition yields

$$0 = \frac{\partial U_i(\theta_i, a)}{\partial a_i} = -2(1+\delta)\Big[(a_i - \theta_i) + \lambda(a_i - \bar{a}_\infty)\Big]. \tag{4}$$

Since U_i is strictly concave, the unique solution to (4) is indeed the global best response of player i of type θ_i. Solving (4) for a_i yields

$$\sigma(\theta_i, \bar{a}_\infty) = \frac{\theta_i + \lambda \bar{a}_\infty}{1+\lambda}.$$

Plugging in the values for θ_i, we obtain

$$\sigma(0, \bar{a}_\infty) = \frac{\lambda \bar{a}_\infty}{1+\lambda}, \qquad \sigma(1, \bar{a}_\infty) = \sigma(0, \bar{a}_\infty) + \frac{1}{1+\lambda}.$$

Since $\bar{a}_\infty$ is the average population action, we know $\bar{a}_\infty = \mu\sigma(1, \bar{a}_\infty) + (1-\mu)\sigma(0, \bar{a}_\infty)$, hence

$$\bar{a}_\infty = \mu\sigma(0, \bar{a}_\infty) + \frac{\mu}{1+\lambda} + (1-\mu)\sigma(0, \bar{a}_\infty) = \sigma(0, \bar{a}_\infty) + \frac{\mu}{1+\lambda} = \frac{\lambda \bar{a}_\infty + \mu}{1+\lambda}.$$

We can solve this identity for $\bar{a}_\infty = \mu$. □

Remark 3.1. Most likely one can show that any Markov strategy has to be monotone. The Markov restriction precludes any punishments based on histories, hence individuals in generation t have to best respond myopically to $\bar{a}_{t-1}$ and $\bar{a}_{t+1}$. If the sensitivity of $\bar{a}_{t+1}$ with respect to $\bar{a}_{t-1}$ is less than one, then the best response is monotone in $\bar{a}_{t-1}$. A formal proof is beyond the scope of this paper.

Given that the average population action converges, we can refine our guess for the simplest Markov equilibrium. Many processes converge geometrically, i.e., $\bar{a}_t = x\bar{a}_{t-1} + (1-x)\mu$ is a convex combination of $\bar{a}_{t-1}$ and the limit μ. More generally, we will characterize all Markov-perfect equilibria in symmetric affine strategies. As it turns out, the only affine Markov-perfect equilibria are convex combinations of the previous generation and the limit.

Theorem 3.2. *For any λ, δ, μ, and $\bar{a}_0$, there exists a unique Markov-perfect equilibrium in symmetric affine strategies, characterized by*

$$\sigma(0, \bar{a}_{t-1}) = \bar{a}_t(\bar{a}_{t-1}) - \frac{\mu}{1+\lambda}, \qquad \sigma(1, \bar{a}_{t-1}) = \bar{a}_t(\bar{a}_{t-1}) + \frac{1-\mu}{1+\lambda}, \tag{5}$$

where $\bar{a}_t(\bar{a}_{t-1})$ *is given by* $\bar{a}_t(\bar{a}_{t-1}) = x\bar{a}_{t-1} + (1-x)\mu$ *for*

$$x = \begin{cases} \frac{(1+\delta)(2+\lambda)-\sqrt{(1+\delta)^2(2+\lambda)^2-4\delta\lambda^2}}{2\delta\lambda} & \text{if } \delta\lambda > 0, \\ \frac{\lambda}{2(1+\delta)+\lambda} & \text{if } \delta\lambda = 0. \end{cases}$$

Given gender norm of either 0 or 1, if social pressure is large, players of a specific type would be forced to act in a similar way to the opposite type. As $\mu = \frac{1}{2}$ by law of large numbers, (5) implies $\sigma(0, \bar{a}_{t-1})$ and $\sigma(1, \bar{a}_{t-1})$ would be equally apart from the average population action in generation t. $\sigma(0, \bar{a}_{t-1})$ is below a_t and $\sigma(1, \bar{a}_{t-1})$ is above a_t. And as λ increases, the difference between $\sigma(0, \bar{a}_{t-1})$ and $\sigma(1, \bar{a}_{t-1})$ would lessen. Therefore, the $\bar{a}_t$ term would be similar to $\bar{a}_{t-1}$. Since the social norm is constituted by the average of the single generation social norm in generations $t-1$ and t, the average population action $\bar{a}_t$ in generation t would play a large role in determining the norm. If the value for $\bar{a}_t$ is similar to that of $\bar{a}_{t-1}$, then b_t would largely be constituted by the $\bar{a}_t$ term. Therefore, if social pressure is large, then the action from generation $t+1$ would not deviate much from past action of generation t. If the action does not change much, then the gender norm in the corresponding generation would also stay relatively constant. Therefore, in the presence of large social pressure, the term $\bar{a}_{t-1}$ would play a greater role than $\bar{a}_\infty = \mu$, and it would take them longer to move away from the initial gender norm which matches the result of Lemma 3.3 (i). In the extreme case where there is no social pressure, then players can act according to their type. Furthermore, with no social pressure, players would have no incentive to comply to the norm set by previous generations. In this case, x would likely be 0.

The δ parameter would have the opposite effect compared to λ. The value of δ affects individual actions in generation t which would then affect the social norm in generation $t+1$. The δ term denotes how important future utility is to the players. Naturally, mothers who are more future oriented would know how their current action might impact the gender norm in period $t+1$ and potentially impact their utility in future generations. If the starting gender norm is opposite to what their type is, then in order to maximize their lifetime utility, mothers would want to change this current social norm. And the intuition behind that is essentially even

if they were not able to act according to their type in the first half of their life, they would want to act more like themselves in the later half of their life. And in order for b_t to change, $\bar{a}_t$ would need to change given they are in generation t. For players who are forward looking enough they would choose $\sigma(\theta_i, \bar{a}_{t-1})$ closer to the value of their own type in order to change $\bar{a}_t$. As they can not change what has already happened ($\bar{a}_{t-1}$), they would focus on changing the future to suit their type. Therefore, they would want to keep changing $\sigma(\theta_i, \bar{a}_t)$ which is similar to what comparative statics suggests: x is decreasing as δ increases. This also implies that they would be moving away from the initial social norm. In the case of $\delta = 0$, players would only care about the utility gained in the first half of their lives. This would be similar to the idea that this player would only live for one generation, which implies they would only need to maximize their utility in generation t. If social pressure of a specific norm ($b_t = 0$) is high in generation t, then type 1 players would act in a similar way to type 0 players as $\sigma(0, \bar{a}_{t-1})$ and $\sigma(1, \bar{a}_{t-1})$ would be similar, and this would be equivalent to looking at how λ affects the equilibrium.

Due to the absence of a prominent player, the results are different to the that of Acemoglu and Jackson (2014). With a continuum of action, the society does not require the signal of a prominent to choose a specific action. Instead, it could itself break away from the gender norm as players in different generations would shift their action $\sigma(\theta_i, \bar{a}_{t-1})$ affecting $\bar{a}_t$ which influences b_t. However, the speed at which the society breaks away from the norm would depend on how large the social pressure is, and how forward looking the players are.

Lemma 3.3. *The Markov-perfect equilibrium has the following monotonicity properties:*

(i) *x is increasing in λ for any value of δ.*

(ii) *x is constant in δ if $\lambda = 0$ and x is decreasing in δ if $\lambda > 0$.*

Because $\bar{a}_t$ is a convex combination of $\bar{a}_{t-1}$ and μ, an increase in x means that the population moves away more slowly from an existing social norm. This means as the social pressure λ increases, the slower the population would be moving away from the existing social norm. If δ gets larger, a_{t-1} will be weighted less. Because $y = 1 - x$, a_∞ will have a larger

weight in a_t. Essentially the more futuristic the players are, the less they would want to care about past actions and the quicker they would want to move towards a_∞.

4 Conclusion

The main contribution of this paper would be showing how a change in a norm set by the society, from past actions, is possible by the shift in action by each player in different generations. A possible extension would be the consideration of having a model where players would live more than 2 generations. This may also have an impact on the speed at which the social norm would change as the average population action of each generation would play less of a part in b_t. It would also be interesting to look at how changing the μ parameter would have an impact on the equilibrium. As having $\mu = \frac{1}{2}$ is a vital factor in concluding $\sigma(0, \bar{a}_{t-1})$ and $\sigma(1, \bar{a}_{t-1})$ are equal distant from the average population action in analyzing how λ would impact the equilibrium. Another extension of the analysis would be showing that the equilibrium obtained in this paper is the unique Markov perfect equilibrium in Markov strategies and not just restricted to affine strategies.

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A Proof of main result

Let us write an arbitrary affine strategy as $\bar{a}_t = x\bar{a}_{t-1} + y\mu$. The proof can be summarized in the following three steps:

(i) We derive the first-order necessary conditions that the individual best responses impose on x and y, and we show that the best-response property implies (5).

(ii) We show that $y = 1 - x$, i.e., the affine strategy is, in fact, a convex combination.

(iii) We check for each candidate values of x obtained in (i) whether the strategy defined in (5) takes values in $[0, 1]$ for any θ_i and $\bar{a}_{t-1}$.

It will turn out that precisely one solution satisfies (iii), giving us existence and uniqueness of a Markov-perfect equilibrium at the same time.

Lemma A.1. *Let $\sigma(\theta_i, \bar{a}_{t-1})$ be a symmetric affine function of $\bar{a}_{t-1}$, giving rise to an average population action $\bar{a}_t = x\bar{a}_{t-1} + y\mu$ such that $\sigma(\theta_i, \bar{a}_{t-1})$ maximizes*

$$U_i(\theta_i, a) = -(1+\delta)(a_i - \theta_i)^2 - \lambda(a_i - b_t)^2 - \delta\lambda(a_i - b_{t+1})^2 \tag{6}$$

if all future generations $s > t$ select $\bar{a}_s$ according to the same affine function. Then $\sigma(\theta_i, \bar{a}_{t-1})$ is determined by (5), x is given by

$$x = \begin{cases} \frac{(1+\delta)(2+\lambda) \pm \sqrt{(1+\delta)^2(2+\lambda)^2 - 4\delta\lambda^2}}{2\delta\lambda} & \text{if } \delta\lambda > 0, \\ \frac{\lambda}{2(1+\delta)+\lambda} & \text{if } \delta\lambda = 0, \end{cases} \tag{7}$$

and for any value of x, y is given by

$$y = \frac{2(1+\delta)}{2(1+\delta)+\lambda(1-\delta x)}. \tag{8}$$

Proof. Because (6) is a concave function of a_i, it is maximized at the critical point which is

$$0 = \frac{\partial U_i(\theta_i, a)}{\partial a_i} = -2(1+\delta)(a_i - \theta_i) - 2\lambda(a_i - b_t) - 2\delta\lambda(a_i - b_{t+1}).$$

Then collecting a_i onto the same side yields the best response of player i

$$2(1+\delta)a_i + 2\lambda a_i + 2\delta\lambda a_i = 2(1+\delta)\theta_i + 2\lambda b_t + 2\delta\lambda b_{t+1}.$$

Solving for a_i yields

$$a_i = \sigma(\theta_i, \bar{a}_{t-1}) = \frac{(1+\delta)\theta_i + \lambda b_t + \delta\lambda b_{t+1}}{(1+\delta)(1+\lambda)}. \tag{9}$$

Let us first derive (5) from (9). To do that, observe that

$$\sigma(1, \bar{a}_{t-1}) = \sigma(0, \bar{a}_{t-1}) + \frac{1}{1+\lambda}. \tag{10}$$

Because $\bar{a}_t$ is the average population action in generation t, another way to write it involving the best responses would be

$$\begin{aligned}
\bar{a}_t &= \mu\sigma(1, \bar{a}_{t-1}) + (1-\mu)\sigma(0, \bar{a}_{t-1}) \\
&= \mu\sigma(0, \bar{a}_{t-1}) + \frac{\mu}{1+\lambda} + (1-\mu)\sigma(0, \bar{a}_{t-1}) \\
&= \sigma(0, \bar{a}_{t-1}) + \frac{\mu}{1+\lambda}.
\end{aligned}$$

Solving this expression for $\sigma(\theta_i, \bar{a}_{t-1})$ yields

$$\sigma(0, \bar{a}_{t-1}) = \bar{a}_t - \frac{\mu}{1+\lambda}. \tag{11}$$

Together with (10), we can deduce

$$\sigma(1, \bar{a}_{t-1}) = \bar{a}_t + \frac{1-\mu}{1+\lambda}. \tag{12}$$

We conclude that in any symmetric affine strategy, best responses by individual players satisfy (5).

$$b_t = \frac{1}{2}(\bar{a}_{t-1} + x\bar{a}_{t-1} + y\mu)$$

and

$$b_{t+1} = \frac{1}{2}(x\bar{a}_{t-1} + y\mu + x(x\bar{a}_{t-1} + y\mu) + y\mu) = \frac{1}{2}(x(1+x)\bar{a}_{t-1} + y(2+x)\mu).$$

Since the expected value of θ_i is μ, it follows from (9) that $\bar{a}_t$ satisfies

$$\bar{a}_t = \frac{\mu}{(1+\lambda)} + \bar{a}_{t-1}\frac{\frac{\lambda}{2}(1+x) + \frac{\lambda\delta}{2}x(1+x)}{(1+\delta)(1+\lambda)} + \mu\frac{\frac{\lambda}{2}y + \frac{\delta\lambda}{2}y(2+x)}{(1+\delta)(1+\lambda)}. \tag{13}$$

We have found two different ways to express $\bar{a}_t$ as a function of $\bar{a}_{t-1}$ and μ: our hypothesis $\bar{a}_t = x\bar{a}_{t-1} + y\mu$ and (13). Because two affine functions coincide if and only if the coefficients of the linear and the constant term agree, we deduce that x and y have to satisfy

$$x = \frac{\frac{\lambda}{2}(1+x) + \frac{\lambda\delta}{2}x(1+x)}{(1+\delta)(1+\lambda)} = \frac{\lambda(1+x)(1+\delta x)}{2(1+\delta)(1+\lambda)} \tag{14}$$

and

$$y = \frac{1}{1+\lambda} + \frac{\lambda y + 2\delta\lambda y + \delta\lambda x}{2(1+\delta)(1+\lambda)}. \tag{15}$$

We first solve for the value of x. Cross-multiplying (14) with the denominator yields

$$2x(1+\delta)(1+\lambda) = \lambda(1+x)(1+\delta x).$$

This is a quadratic equation in x that we can bring into standard form

$$-\lambda\delta x^2 + (\lambda+2)(\delta+1)x - \lambda = 0. \tag{16}$$

If $\delta\lambda = 0$, then this is in fact a linear equation with the unique solution

$$x = \frac{\lambda}{(1+\delta)(2+\lambda)} = \frac{\lambda}{2(1+\delta)+\lambda}, \tag{17}$$

where we have used the fact that $\delta\lambda = 0$ to simplify the denominator. If $\delta\lambda \neq 0$, the quadratic

formula yields two possible solutions x_+ and x_-, given by (7). Luckily, (15) is linear in y. Multiplying (15) with $2(1+\lambda)(1+\delta)$ yields

$$2(1+\delta)(1+\lambda)y - 2(1+\delta) = \lambda y\Big(1+\delta(2+x)\Big).$$

Solving for y yields the solution (8). This concludes the proof. □

Lemma A.2. *For any values of x and y given in Lemma A.1, we have $y = 1 - x$.*

Proof. Consider first the case $\delta\lambda = 0$. Evaluating (8) yields

$$y = \frac{2(1+\delta)}{2(1+\delta)+\lambda} = 1 - x. \tag{18}$$

For the case $\delta\lambda > 0$, we first check monotonicity of y. From the derivative of y with respect to x

$$\frac{\partial y}{\partial x} = \frac{2\delta\lambda(1+\delta)}{(2(1+\delta)+\lambda(1-\delta x))^2} > 0$$

for all values of x because $\delta, \lambda > 0$ implies numerator is positive and the denominator is positive. Since y is monotonically increasing, it would be a 1-1 mapping between x and y, i.e., $y(x)$ is the unique value that satisfies (8). Let us show that $1 - x$ also equals the right-hand side of (8) to deduce that $y = 1 - x$. Equating $y(x) = 1 - x$ and multiplying both sides with $2(1+\delta) + \lambda(1-\delta x)$ yields

$$2(1+\delta) = (1-x)(2(1+\delta)+\lambda(1-\delta x)).$$

With further simplification, the equation would be the same as (16), implying that $x_\pm$ are precisely the two values, for which $y(x_\pm) = 1 - x_\pm$. □

Lemma A.3. *There exist a unique MPE in affine strategies if $\delta\lambda = 0$.*

Proof. We want to check the x and y components for $\bar{a}_t$ yields a valid strategy. A valid strategy requires $\bar{a}_t \in [0, 1]$. the lower bound for $\bar{a}_t$ using (11) and (12) is

$$\frac{\mu}{1+\lambda}$$

and the upper bound for $\bar{a}_t$ is

$$1 - \frac{1-\mu}{1+\lambda}$$

For the $\delta = 0$, we first check $\bar{a}_t(0) > \frac{\mu}{1+\lambda}$.

$$\bar{a}_t(0) = \frac{2\mu}{2+\lambda} = \frac{\mu}{1+\frac{\lambda}{2}}$$

Because $1 + \lambda > 1 + \frac{\lambda}{2}$, we can conclude

$$\frac{\mu}{1+\lambda} < \frac{\mu}{1+\frac{\lambda}{2}}.$$

For the upper bound, we want to check

$$\frac{\lambda}{\lambda+2} + \frac{2\mu}{2+\lambda} \overset{?}{<} 1 - \frac{1-\mu}{1+\lambda}.$$

$$\frac{\lambda+2\mu}{2+\lambda} \overset{?}{<} \frac{\lambda+\mu}{1+\lambda}$$

$$\lambda + \lambda^2 + 2\mu + 2\mu\lambda \overset{?}{<} 2\lambda + \lambda^2 + 2\mu + \mu\lambda$$

$$\lambda(1+2\mu) \overset{?}{<} \lambda(2+\mu)$$

$$1 + 2\mu \overset{?}{<} 2 + \mu$$

Because $\mu \in (0,1)$, values of $1 + 2\mu \in (1,3)$ and values of $2 + \mu \in (2,3)$, and we therefore conclude

$$\frac{\lambda+2\mu}{2+\lambda} \leq \frac{\lambda+\mu}{1+\lambda}.$$

Since $\bar{a}_t$ lies between the lower and upper bound, this is a valid strategy.

For $\lambda = 0$, we know $x = 0$ and $y = 1$ from lemma 4.2. Therefore, $\bar{a}_t = \mu$ which is in [0,1], which is a valid strategy. □

Lemma A.4. *There exist a unique MPE in affine strategies for the $\delta\lambda > 0$ case.*

We know x and y values from (7) and (8). If

$$x = \frac{(\lambda+2)(\delta+1) - \sqrt{(\lambda+2)^2(1+\delta)^2 - 4\delta\lambda^2}}{2\delta\lambda}$$

then this would not yield a valid strategy as $x > 1$ and this makes $\bar{a}_t > 1$. We will show this by showing the numerator is always greater than the denominator for all values of δ and λ. If $\lambda \in [0, \infty)$, with $\delta \in \{\frac{a}{b} \mid a, b \in \mathbb{Z}_+ \text{ and } a < b\}$. Since we know $\sqrt{(\lambda+2)^2(1+\delta)^2} > 0$, we only need to check if $b(\lambda+2)(\frac{a}{b}+1) > 2a\lambda$.

If $b < 2a\lambda$, then the first term of the numerator is

$$(b\lambda + 2b)(\frac{a}{b} + 1)$$

$$\lambda a + b\lambda + 2a + 2b.$$

Since $b > a$, then $\lambda(a+b) > \lambda(a+a) = 2\lambda a$ and we have shown that the numerator is greater than the denominator. For the other value of x, we want to check if

$$\frac{\mu}{1+\lambda} < \bar{a}_t < 1 - \frac{1-\mu}{1+\lambda}$$

We first check whether $\bar{a}_t(0) > \frac{\mu}{1+\lambda}$. Because $\sqrt{a-b} \geq \sqrt{a} - \sqrt{b}$

$$x \leq \frac{2\lambda\sqrt{\delta}}{2\delta\lambda} = \frac{1}{\sqrt{\delta}}.$$

Therefore, we know that $x = 1-y < \frac{1}{\sqrt{\delta}} < 1$ and $y > 0$. From this, we also know that $1-\delta x < 1$ which makes the denominator of y less than that of $\frac{2(1+\delta)}{2(1+\delta)+\lambda}$ (the y value for $\delta\lambda = 0$). Since

$$\frac{2\mu(1+\delta)}{2(1+\delta)+\lambda} \geq \frac{\mu}{1+\lambda},$$

we conclude that $\bar{a}_t(0) = \mu y \geq \frac{2\mu(1+\delta)}{2(1+\delta)+\lambda}$ which is greater than the lower bound.

B Proof of Lemma 3.3

For the case of $\delta\lambda = 0$ if $\delta = 0$, then the derivative of x with respect to λ is

$$\frac{\partial x}{\partial \lambda} = \frac{(\lambda+2) - \lambda}{(\lambda+2)^2} > 0.$$

If there is no social pressure, $\lambda = 0$, then $\bar{a}_t = a_\infty$. Because $a_\infty = \mu$ when there is no social pressure, players would act according to their type.

For the case of $\delta\lambda > 0$, we will be looking at the x value of

$$x = \frac{(\lambda+2)(\delta+1) - \sqrt{(\lambda+2)^2(\delta+1)^2 - 4\delta\lambda^2}}{2\delta\lambda}.$$

We will first look at how δ affects x, which is done through taking the derivative of x with respect to δ. We first let $u(\delta)$ be

$$u(\delta) = (\lambda+2)(\delta+1) - \sqrt{(\lambda+2)^2(1+\delta)^2 - 4\delta\lambda^2}.$$

And let

$$v(\delta) = \sqrt{(\lambda+2)^2(1+\delta)^2 - 4\delta\lambda^2}.$$

Then the derivative of $u(\delta)$ with respect to δ is

$$\frac{\partial u(\delta)}{\partial \delta} = \lambda + 2 - v'(\delta)$$

where $v'(\delta)$ is

$$\begin{aligned}\frac{\partial v(\delta)}{\partial \delta} &= \frac{1}{2}((\lambda+2)^2(1+\delta)^2 - 4\delta\lambda^2)^{\frac{-1}{2}}(\frac{\partial}{\partial \delta}[(\lambda+2)^2(1+\delta)^2 - 4\delta\lambda^2] \\ &= \frac{2(\lambda+2)^2(\delta+1) - 4\lambda^2}{2\sqrt{(\lambda+2)^2(1+\delta)^2 - 4\delta\lambda)}}.\end{aligned}$$

Then the derivative of x with respect to δ is

$$\frac{\partial x}{\partial \delta} = \frac{2\delta\lambda(\lambda+2-v'(\delta)) - 2\lambda u(\delta)}{4\delta^2\lambda^2}.$$

Then we want to check the sign of the derivative of x with respect to δ. Since the denominator is positive, we only need to check the sign of the numerator. We know that $2\delta\lambda < 2\lambda$, and then we need to check the relationship between $\lambda+2-v'$ and $u(\delta)$. Because both $\lambda, \delta \geq 0$, with $\lambda + 2 < (\lambda+2)^2(\delta+1)^2$ then check which of $v'(\delta)$ or v is larger. To make the process

easier, multiply both $v'(\delta)$ and v with $\sqrt{(\lambda+2)^2(\delta+1)^2-4\delta\lambda^2}$. Then we are comparing

$$(\lambda+2)^2(\delta+1)-4\lambda^2$$

with

$$(\lambda+2)^2(\delta+1)^2-4\delta\lambda^2.$$

We subtract the second term from the first to see which one is larger

$$(\lambda+2)^2(\delta+1)-4\lambda^2-(\lambda+2)^2(\delta+1)^2+4\delta\lambda^2.$$

The terms rewritten gives

$$(\lambda+2)^2(\delta+1)(-\delta)+4\lambda^2(\delta-1)<0.$$

The first term is always negative and the sign of the second term is always negative as well. This implies $v'(\delta)<v(\delta)$. Then we want to check which of

$$\lambda+2-v'(\delta)$$

and

$$(\lambda+2)^2(\delta+1)^2-v(\delta)$$

is larger. Subtracting the second term from the first, we get

$$(\lambda+2)(1-(\lambda+2)(\delta+1)^2)+(v-v').$$

The first term is negative and the second term is positive. The value of the second term is less than the first term which leaves the numerator to be negative. Therefore, $\frac{\partial x}{\partial \delta}<0$.

Finally, let us take the derivative of x with respect to λ. To make the derivatives more manageable, abbreviate $a=2(1+\delta)$ and

$$g(\lambda)=a^2(2+\lambda)^2-4\delta\lambda^2, \qquad g'(\lambda)=2a^2(2+\lambda)-8\delta\lambda=\frac{2g(\lambda)-4a^2(2+\lambda)}{\lambda}. \tag{19}$$

The quotient rule implies that

$$x'(\lambda) = \frac{2\delta\lambda\left[a - \frac{g'(\lambda)}{2\sqrt{g(\lambda)}}\right] - 2\delta\left[a(2+\lambda) - \sqrt{g(\lambda)}\right]}{4\delta^2\lambda^2} = \frac{\sqrt{g(\lambda)} - 2a - \frac{\lambda g'(\lambda)}{2\sqrt{g(\lambda)}}}{2\delta\lambda^2}.$$

Using (19), we can simplify the term

$$\frac{\lambda g'(\lambda)}{2\sqrt{g(\lambda)}} = \frac{2g(\lambda) - 4a^2(2+\lambda)}{2\sqrt{g(\lambda)}} = \sqrt{g(\lambda)} - \frac{2a^2(2+\lambda)}{\sqrt{g(\lambda)}}.$$

The $\sqrt{g(\lambda)}$ terms will cancel out. After factoring out $2a/\sqrt{g(\lambda)}$, we obtain

$$x'(\lambda) = \frac{2a\left(a(2+\lambda) - \sqrt{g(\lambda)}\right)}{2\delta\lambda^2\sqrt{g(\lambda)}} = \frac{2ax_-(\lambda)}{\lambda\sqrt{g(\lambda)}} > 0.$$